

TOKENISED DEPOSITS, CENTRAL-BANK MONEY AND CONTROL

Project Agorá

A tour of the eight Agorá currency areas and the architecture they are testing

Integrated briefing current to 5 August 2026

What Project Agorá is

- A BIS Innovation Hub / IIF public–private experiment: tokenised **commercial-bank deposits** and tokenised **central-bank reserves** on one programmable platform.
- Seven founding currency areas – euro, yen, won, peso, franc, sterling, dollar – with their central banks; **Canada joined May 2026**.
- Over forty private financial firms as design and testing participants.
- Findings report 27 May 2026; real-value transactions executed July 2026.

Tokenised commercial-bank deposits

↓ shared payment contracts and atomic conditions ↓

Separate currency-specific central-bank reserve domains

Sources: BIS, Agorá findings report (27 May 2026); BIS, real-value testing summary; Bank of Canada accession release, 27 May 2026.

From dollar to ledger

Formal authority

- Central bank issues the settlement asset
- Domestic law defines finality
- RTGS records the final debit and credit

Operational authority

- Identity and admission
- Routing, sequencing and liquidity locks
- Code, keys, data, recovery and exit

Sources: RBA, [RITS overview](#); Bank of England, [RTGS overview](#).

Vocabulary

RTGS	real-time gross settlement: payments settle one by one, immediately, in central-bank money (Fed-wire, T2, BOJ-NET, RITS)
DNS	deferred net settlement: obligations accumulate and net positions settle in batches at designated times
CSD	central securities depository: the definitive register of securities ownership; transfers become book entries
CCP	central counterparty: interposes itself in every trade by novation; manages credit risk through margin and default funds
Finality	the state in which fulfilled obligations are discharged irrevocably – technical (no internal reversal) and legal (no external reversal)

Finality releases liquidity. Liquidity bounds finality. Programmable settlement changes both.

Sources: CPMI-IOSCO PFMI; definitions adapted from the companion Liquidity Automata research.

Orders of magnitude

- **Swift** moves value equivalent to world GDP roughly every three days.
- **CLS** settled more than US\$8 trillion of FX on an average day in 2025, with a US\$19.1 trillion record day.
- **LCH SwapClear** clears about 95% of vanilla OTC interest-rate swaps globally, in 28 currencies.
- **Kinexys** (J.P. Morgan) reports over US\$4 trillion processed since inception, now in eight currencies.

These layers govern flow, not stock. Small design choices compound across these volumes daily.

Sources: Swift annual review; CLS 2025 averages; LSEG SwapClear; J.P. Morgan Kinexys milestones (July 2026).

Control of a settlement system is an eight-layer capability

-
- | | | | |
|---|-------------------------------------|---|---|
| 1 | Law and finality | 5 | Code, keys and specialist knowledge |
| 2 | Rules, access and pricing | 6 | Connectivity, identity and certificates |
| 3 | Authoritative money or title record | 7 | Data, routing and optimisation |
| 4 | Live operations and recovery | 8 | Credible exit and substitution |

A jurisdiction can retain layers 1–3 while conceding layers 4–8 to external vendors and international utilities.

Layer tags **L1–L8** recur on the jurisdiction and dependency tables that follow.

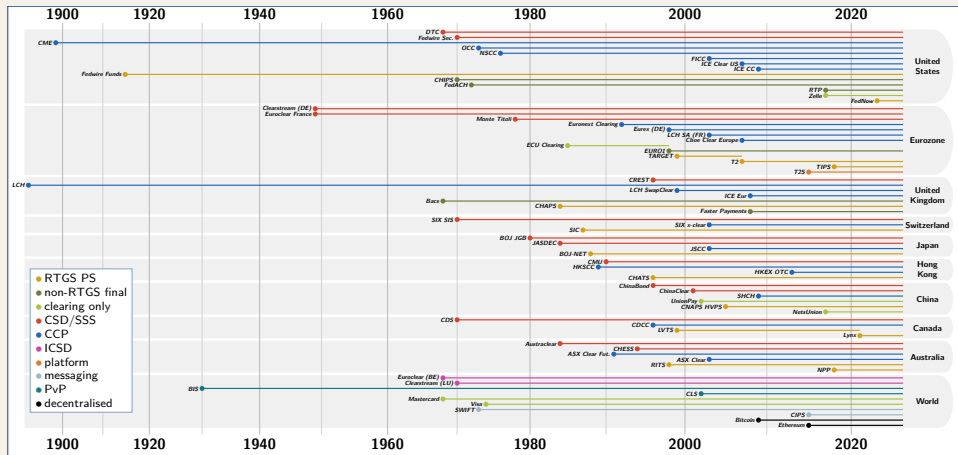
The central risk is a private constitution

A shared technical layer can become the place where market access, liquidity and permissible behaviour are actually governed.

- No conspiracy is required: institutions rationally join every credible network.
- Network effects make the chosen identity, token and messaging standards difficult to escape.
- The result can resemble coordination because the same firms occupy every architecture.

Sources: Ownership and participation analysis in the companion report; public corporate and FMI disclosures.

A century of financial market infrastructure



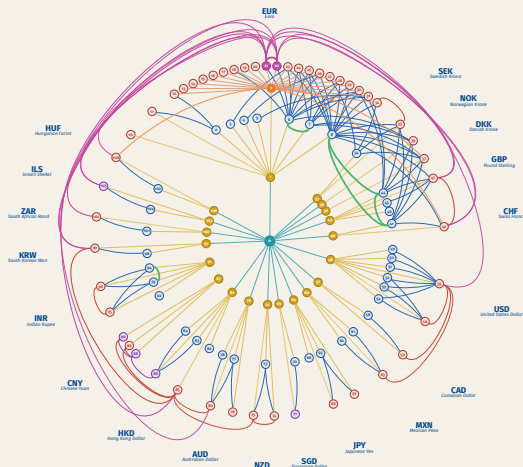
Sources: FMI timeline from the companion Liquidity Automata research (Kelly, UNSW).

PART I

Round the world: the eight Agorá jurisdictions

Tokenised deposits, central-bank money and the programmable transition

The incumbent core is already a network



Node and link key

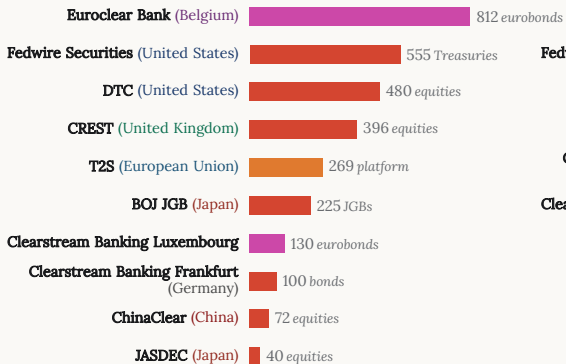
- Continuous Linked Settlement (PvP)
- Real-Time Gross Settlement (RTGS)
- Central Counterparty (CCP)
- Central Securities Depository (CSD)
- International Central Securities Depository (ICSD)
- Central Securities Depository / Central Counterparty
- Eurosystem Platform (TARGET2-Securities)

- Continuous Linked Settlement
- Payment Settlement
- Securities Clearing
- CSD-CSD Link
- TARGET2-Securities Settlement
- ICSD-CSD Link
- ICSD-ICSD (The Bridge)
- CCP Interoperability Link

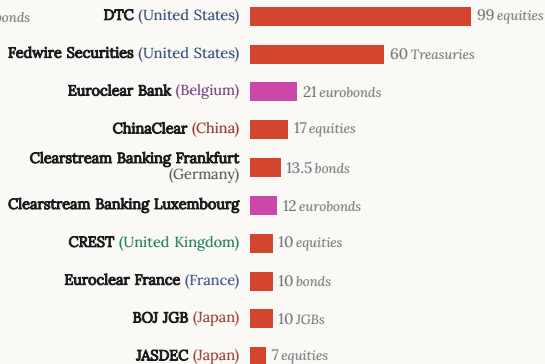
Sources: Global FMI network map from the companion Liquidity Automata research (Kelly, UNSW).

Major CSDs

annual deliveries value, \$ T

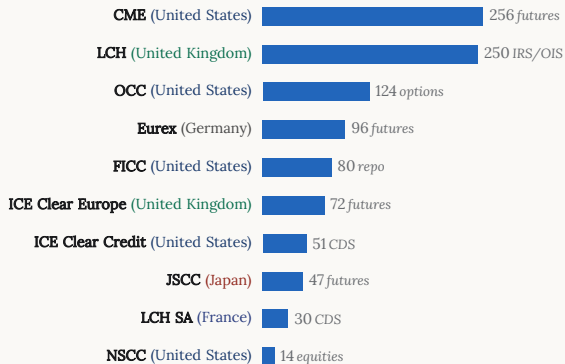


assets under custody, \$ T

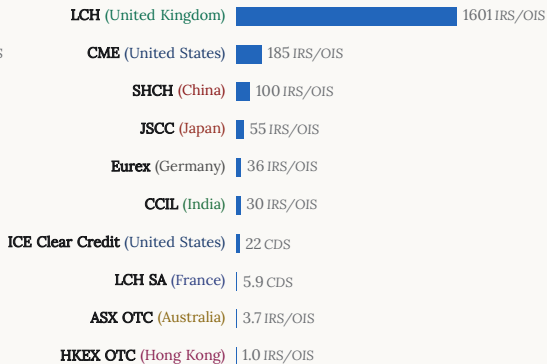


Major CCPs

initial margin held, \$ B

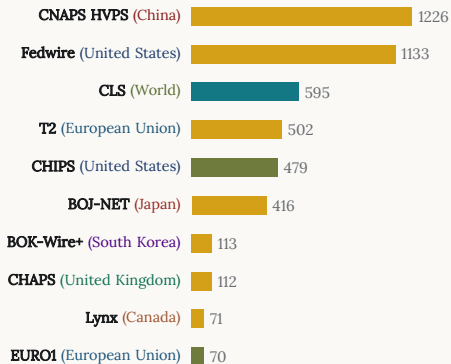


annual OTC notional cleared, \$ T

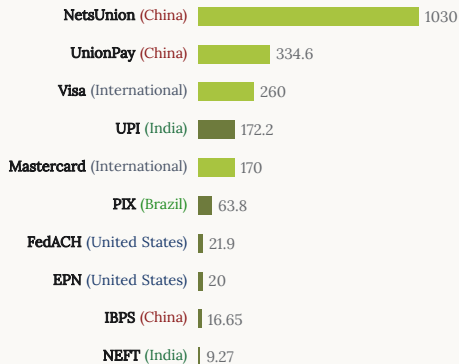


Major Payment Systems

annual settlement value, \$T



annual volume, billions of txns



Payments | two public settlement services, four private schemes

T2

Eurosystem

RTGS for euro high-value payments and the cash accounts used by ancillary systems.

Central-bank money; final on account entry; open each TARGET business day.

EURO1 / STEP2

EBA Clearing

Private large-value net settlement and pan-European retail batch clearing.

EURO1 uses a single obligation structure; both settle final positions in TARGET.

TIPS

Eurosystem

24/7 instant settlement for SCT Inst payments, reached through participant and reachable-party accounts.

Prefunded central-bank money; transaction-by-transaction finality.

RT1

EBA Clearing

24/7 pan-European instant-payment infrastructure for SCT Inst.

Private scheme with prefunded settlement positions in TIPS.

Securities FMIs | settlement is shared; custody and clearing are not

T2S

Eurosystem platform

Common technical platform on which connected CSDs settle securities against dedicated cash accounts.

DvP in central-bank money; CSDs retain their legal relationships and rulebooks.

ICSDs

Euroclear Bank; Clearstream Banking Luxembourg

Cross-border custody, settlement and collateral services, including international bonds.

Euroclear Bank cash is commercial-bank money; Clearstream also connects to central-bank rails.

National CSDs

Euroclear France; Euronext Securities Milan;

Iberclear

Issue, safekeep and settle domestic securities while outsourcing the settlement engine to T2S.

National legal records above a common Eurosystem platform.

CCPs

Eurex; LCH SA; Euronext Clearing; BME; Cboe Clear Europe

Novate and net equity, bond, repo and derivatives exposures; collect margin and manage defaults.

Private risk engines feeding cash and securities settlement systems.

Digital money | retail CBDC and wholesale DLT are separate programmes

RETAIL CBDC

Digital euro

A digital form of central-bank money for households and firms. The ECB has made no issuance decision; legislation and a later Governing Council decision remain prerequisites.

Purpose: public money in an increasingly digital retail economy; possible first issuance from 2029 if approved.

WHOLESALE END-STATE

Appia

Longer-term integrated ecosystem for wholesale central-bank-money settlement of tokenised assets. A blueprint is targeted for 2028.

Purpose: move from bilateral bridges to a common European wholesale architecture.

WHOLESALE BRIDGE

Pontes

Interim Eurosystem service connecting market DLT platforms to TARGET Services for settlement in central-bank money. Pilot planned from Q3 2026.

Purpose: a central-bank cash leg for tokenised securities without replacing T2/T2S.

CROSS-BORDER

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Eight euro-area banks tested tokenised deposits against tokenised reserves on a shared cross-currency workflow.

Purpose: correspondent payment and PvP, not retail CBDC and not securities issuance.

Payments | the central bank and Zengin split wholesale from retail

BOJ-NET Funds Transfer

Bank of Japan

RTGS for money-market, FX, securities and customer payments between account holders.

Central-bank money; liquidity-saving features queue and offset eligible payments.

Zengin System

Japanese Banks' Payment Clearing Network

Domestic credit transfers between deposit-taking institutions; gateway for customer bank transfers.

Multilateral net obligations settle through BOJ-NET; more-time system supports 24/7 messages.

CLS yen leg

CLS Bank with BOJ-NET

PvP settlement of yen against other eligible currencies during the common settlement window.

Yen funding and payouts pass through BOJ-NET accounts.

Retail overlays

Banks and payment providers

Cards, QR wallets and funds-transfer services sit above bank accounts and the Zengin/BOJ-NET core.

They change initiation and access, not the final settlement asset.

Securities FMI | government bonds and private securities take different routes

BOJ-NET JGB Services

Bank of Japan

Book-entry transfer and DvP settlement for Japanese government bonds.

Securities and cash legs settle on Bank of Japan infrastructure.

JSCC

Japan Exchange Group

CCP for cash equities, listed derivatives, OTC interest-rate swaps and government-bond transactions.

Novation, multilateral netting, margin and default management before DvP.

JASDEC

Market-owned CSD

Book-entry transfer for equities, corporate bonds, investment trusts and other private instruments.

Links securities settlement to cash-settlement banks and BOJ-NET arrangements.

Token venues

BOOSTRY / ibet for Fin; ODX START

Permissioned security-token registry and trading venue used for digital corporate bonds and secondary transactions.

DCJPY supplies a programmable deposit-money cash leg; it does not clear the security itself.

Digital money | CBDC research and DCJPY serve different markets

RETAIL CBDC RESEARCH

BOJ CBDC pilot

The Bank of Japan is testing an end-to-end retail-CBDC system with intermediaries and an external expert forum. It has made no decision to issue.

Purpose: test performance, resilience, intermediary functions and coexistence with cash and bank deposits.

ENERGY + SECURITIES

What DCJPY settles

First production use paid for digital environmental-value certificates. In 2026 escrow DvP linked DCJPY to tokenised corporate bonds and secondary-market security tokens.

Purpose: eliminate principal risk and manual reconciliation between a token security and its bank-money cash leg.

TOKENISED DEPOSIT

DCJPY

A bank's yen deposit liability represented on DCP's permissioned two-zone network. GMO Aozora began production issuance in August 2024.

Financial Zone mints and burns against core deposits; Business Zones add escrow and application logic.

CROSS-BORDER

Project Agorá

Mizuho, MUFG, SBI Shinsei and SMBC tested yen deposit tokens against tokenised central-bank reserves.

Purpose: cross-border bank payments and FX PvP; distinct from both the retail CBDC pilot and domestic securities DvP.

Payments | a hybrid RTGS above dense retail clearing

BOK-Wire+

Bank of Korea

Hybrid high-value system combining RTGS with simultaneous bilateral and multilateral offsetting.

Central-bank money; intraday liquidity and queuing reduce gridlock.

Electronic Banking System

Korea Financial

Telecommunications & Clearings Institute

24/7 online and mobile interbank transfers; Korea's principal retail faster-payment network.

Transactions are immediate to users; interbank net obligations settle later in BOK-Wire+.

CLS won leg

CLS Bank with BOK-Wire+

PvP settlement of eligible won FX trades.

Funding and payout connect the global FX utility to the domestic public ledger.

Other retail systems

KFTC

CD/ATM, giro, cheque, cash-management and local-bank networks.

KFTC clears; the Bank of Korea settles final interbank positions.

Securities FMIs | KRX clears; KSD records and settles

Korea Securities Depository

Market utility

CSD, book-entry transfer and securities settlement for exchange and OTC markets.

DvP cash obligations settle through designated banks or BOK-Wire+, depending on market and participant.

KRX cash markets

Korea Exchange

Exchange, market operator and CCP for listed equities and bonds.

KRX novates and nets trades; KSD completes securities settlement.

OTC bond settlement

KSD + BOK-Wire+

Institutional bond trades are matched and settled securities-against-funds.

Central-bank-money DvP is available to eligible participants.

KRX derivatives / OTC

Korea Exchange

CCP for listed derivatives and eligible OTC won interest-rate swaps.

Margin and default resources sit at KRX; payments settle through banking and BOK rails.

Digital money | Project Hangang rehearses a two-tier token system

Wholesale CBDC

BANK SETTLE-
MENT ASSET

The Bank of Korea issues a tokenised settlement asset only to participating institutions on the shared ledger.

Purpose: keep central-bank money at the base of tokenised bank liabilities.

Who and how

BOK + NINE BANKS

BOK runs the core ledger; banks handle customer wallets, KYC and redemption. Smart-contract conditions can restrict where a grant is spent.

Technology: shared DLT with wholesale-CBDC settlement beneath deposit tokens.

Deposit tokens

CUSTOMER MONEY

Participating banks issue programmable claims to customers. Phase 1 ran April to June 2025 at real merchants; Phase 2 launched in March 2026.

Purpose: retail payment and conditional public-grant disbursement, with wallets and merchant QR acceptance.

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CROSS-BORDER

Hana, IBK, KB Kookmin, NongHyup, Shinhan and Woori appear in both the domestic and Agorá experiments.

Agorá adds cross-currency path discovery and PvP; Hangang tests won use cases inside Korea.

Payments | SPEI is more than a retail transfer app

SPEI

Banco de México

Systemically important electronic transfer system for bank, government, financial-market and customer payments.

Liquidity-saving hybrid design; final and irrevocable when settlement notices are issued.

Retail access to SPEI

Banks and non-banks

Customer interfaces accept 24/7 transfer instructions even though the core settlement day has defined windows.

CoDi adds QR/NFC payment requests; it moves ordinary account money and is not a CBDC.

SIAC

Banco de México

Master-account, liquidity and settlement service for banks and public-sector account holders.

Provides central-bank account and liquidity functions used around other payment systems.

SPID / CLS

Banxico and CLS

SPID handles domestic US-dollar transfers between Mexican legal entities; CLS provides PvP for eligible peso FX.

Separate rails for domestic dollar payments and global FX settlement.

Securities FMIs | Indeval links DALI directly to central-bank cash

Indeval

BMV Group CSD

Central securities depository for government and private securities; maintains book-entry ownership records.

Private operator under public regulation.

CCV

BMV Group CCP

Central counterparty for exchange-traded equities.

Novates and nets obligations before DALI settlement.

DALI

Indeval

Real-time securities settlement using DvP and optimisation of linked cash and securities instructions.

Cash accounts are held at Banco de México; Indeval grants no settlement credit.

Asigna

MexDer group trust

CCP for listed derivatives traded on MexDer.

Collects margin, manages default exposure and settles through designated financial institutions.

Digital money | account-money innovation and Agorá

CoDi

PAYMENT OVERLAY

QR and NFC payment requests routed through SPEI. The payer and payee still use conventional bank or e-money accounts.

Purpose: low-cost retail acceptance and financial inclusion; it is neither tokenised money nor a digital peso.

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WHOLESALE CROSS-BORDER

Banxico and BBVA México, Banorte and Monex tested tokenised reserves and bank deposits for wholesale cross-border payments.

Purpose: pathfinding, compliance preparation and atomic multi-ledger settlement for correspondent payments and FX.

Retail CBDC

NO CURRENT IS- SUANCE DECISION

Banco de México has not announced a live retail-CBDC pilot or an issuance decision. Earlier policy references do not amount to an operating system.

This absence matters: do not relabel Agorá or CoDi as Mexico's retail CBDC.

Policy question

ANCHOR + ACCESS

The prototype keeps the peso settlement asset under Banxico while placing workflow and cross-border access on a shared platform.

The unresolved issue is the production operator, rulebook and fallback if that shared layer is unavailable.

Payments | SIX operates the rail; the SNB supplies final money

Swiss Interbank Clearing

SIX Interbank Clearing under SNB

mandate

Near-continuous RTGS for high-value and retail franc payments, including instant payments.

Final and irrevocable settlement in sight deposits at the SNB.

euroSIC

SIX Interbank Clearing

Swiss euro payment system linked to European payment arrangements.

Euro settlement through the system's banking and TARGET connections, not SNB-issued money.

SIC instant payments

SIX / SNB

24/7 instant settlement service introduced in 2024, reached through participating banks.

Same central-bank-money anchor, new continuous service window.

TWINT

Bank-owned wallet company

QR and app-based retail payments linked to bank accounts and cards.

Access and messaging layer; underlying obligations ultimately settle through bank rails.

Securities FMIs | SIX spans exchange, CSD, CCP and digital venue

SIX SIS

SIX Group

Swiss CSD, custody and settlement for domestic and international securities.

DvP through linked cash accounts and SIC arrangements.

SIX Digital Exchange

SIX Group

Regulated exchange and CSD for natively issued digital securities.

Production venue used by Project Helvetia for on-ledger central-bank-money DvP.

SIX x-clear

SIX Group

CCP for Swiss and selected European cash-equity markets.

Interoperates with other European CCPs; manages margin and defaults.

BX Digital link

BX Swiss + SIC

Alternative model in which tokenised securities settle while cash remains on conventional SIC accounts.

A synchronisation mechanism joins the DLT asset leg to the public-money cash leg.

Digital money | Helvetia answers the cash-leg question directly

Helvetia wCBDC

LIVE WHOLESALE PILOT

The SNB issues tokenised central-bank money onto SDX for eligible institutions. It has settled tokenised bonds and SNB bills in production.

Purpose: risk-free DvP for natively digital securities; not a retail franc.

Who controls what

SNB + SIX

SNB controls issuance and redemption; SIX operates SIC, SDX and much of the securities stack.

The public anchor remains, but operational concentration sits at the platform group.

RTGS link

ALTERNATIVE
ARCHITECTURE

A trigger synchronises DLT securities with conventional SIC cash rather than placing wCBDC on the venue.

Purpose: retain the existing ledger while still enabling atomic-style DvP.

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CROSS-BORDER

UBS and Sygnum joined the bank cohort; SDX supported later testing.

Purpose: extend the two-tier token model from Swiss securities DvP to cross-border deposits and FX settlement.

Payments | RT2 settles; schemes determine the flow

RT2 / CHAPS

Bank of England

RTGS infrastructure and high-value same-day sterling payment system.

Direct settlement in reserves; CHAPS payments are final and irrevocable.

Faster Payments

Pay.UK

24/7 near-real-time retail credit transfers, including standing-order and account-to-account services.

Scheme obligations settle on a deferred-net basis through RTGS.

CREST cash link

Bank of England + Euroclear UK &

International

RTGS accounts provide the sterling cash leg for securities DvP in CREST.

Central-bank money is synchronised to securities movements.

Bacs

Pay.UK

Three-day bulk cycle for direct credits and direct debits.

Multilateral net settlement in Bank of England money.

Securities FMIs | a foreign-owned CSD and three specialist CCP groups

CREST

Euroclear UK & International

UK CSD for equities, gilts and money-market instruments; also registrar and corporate-action infrastructure.

DvP cash settles across Bank of England accounts through the CREST/RTGS link.

ICE Clear Europe

Intercontinental Exchange

CCP for energy, credit and financial derivatives.

Commercial settlement banks and central-bank facilities support cash flows.

LCH

London Stock Exchange Group

CCP for rates, repo, credit, FX and equities, including SwapClear and RepoClear.

Global private clearing utility with margin, netting and default-management power.

LME Clear

Hong Kong Exchanges and Clearing group

CCP for London Metal Exchange contracts.

Vertically integrated exchange and clearing chain for base metals.

Digital money | four projects, four different liabilities

Digital pound

RETAIL CBDC RESEARCH

A potential Bank of England liability for households and firms, distributed through private wallet providers. No issuance decision has been made.

Design phase ends in 2026; Parliament and government would be involved before any launch.

GBTD

TOKENISED BANK DEPOSITS

UK-bank work on interoperable digital representations of ordinary commercial-bank deposits.

Purpose: keep bank money usable across programmable ledgers without converting every payment to stablecoin or CBDC.

Finality GBP

RESERVE-BACKED SYSTEM MONEY

Wholesale token backed by funds in an omnibus Bank of England account, used only inside Finality's regulated payment system.

Purpose: 24/7 programmable settlement for wholesale tokenised assets and payments.

Synchronisation / wCBDC

WHOLESALE EXPERIMENTS

Bank of England tests link-based settlement and possible on-ledger central-bank money for DvP and cross-ledger transactions.

Agorá adds the multi-currency correspondent-payment problem.

Payments | competing public and bank-owned rails

Fedwire Funds

Federal Reserve Banks

RTGS for high-value and time-critical payments.

Each transfer is final when credited in central-bank money.

CHIPS

The Clearing House

Large-value dollar system using prefunding, matching and offsetting to conserve liquidity.

Bank-owned private rulebook; final positions backed by Federal Reserve account money.

FedNow / FedACH / NSS

Federal Reserve Banks

24/7 instant settlement, retail ACH processing and multilateral settlement for private networks.

Three public services with different speed, credit and liquidity models.

RTP / EPN

The Clearing House

24/7 instant retail payments and private ACH processing.

Private alternatives to FedNow and FedACH, owned by major banks.

Securities FMIs | DTCC dominates post-trade; CCPs remain specialised

DTC

DTCC subsidiary

CSD and settlement for corporate and municipal securities; receives netted obligations from NSCC.

Net cash settles through designated commercial settlement banks.

NSCC / FICC

DTCC subsidiaries

CCPs for equities, government securities and mortgage-backed securities.

Novation, netting, margin and default management before settlement.

Fedwire Securities

Federal Reserve Banks

Book-entry custody and transfer for Treasuries, agencies and certain international organisations.

DvP in central-bank money over Fedwire.

CME / OCC / ICE

Clearing groups and member utility

CCPs for futures, options, rates, commodities, equities and credit.

Specialised private risk engines above public-dollar and commercial-bank cash settlement.

Digital money | private token dollars and reserve-based public money

US CBDC

NO ISSUANCE
PROGRAMME

The Federal Reserve researches digital money but has not created a retail or wholesale CBDC platform; issuance would require clear legal and political authority.

The dollar's public digital settlement asset remains reserve balances on existing Federal Reserve systems.

Kinexys / JPMD

BANK DEPOSIT IN-
FRASTRUCTURE

JPMorgan's permissioned ledger and deposit-token services move institutional cash and support delivery-versus-payment workflows.

Liability remains with the issuing bank; access is contractual and permissioned.

TCH deposit token

BANK-OWNED INITIATIVE

Shared-ledger tokenised deposits proposed by The Clearing House and member banks for round-the-clock wholesale payments and programmable settlement.

Purpose: preserve regulated bank money while extending service beyond legacy windows.

Stablecoins + Agorá

SEPARATE MODELS

Stablecoins are non-bank or specially regulated issuer liabilities. Agorá instead tests tokenised bank deposits against tokenised central-bank reserves.

US cohort: JPMorgan, Citi, BNY, FNBO and TD Bank N.A.; Visa, Mastercard and Swift also participated.

Payments | Lynx is live; the retail real-time rail is still arriving

Lynx

Payments Canada

Systemically important high-value payment system with RTGS and a liquidity-saving mechanism.

Final settlement in Bank of Canada money; risk controls and collateral support intraday liquidity.

Interac e-Transfer

Interac

Near-real-time retail account-to-account service reached through participating financial institutions.

User messaging is immediate; underlying interbank positions feed established settlement arrangements.

ACSS

Payments Canada

Batch clearing for cheques, pre-authorised debits, POS and other retail instruments.

Multilateral net obligations settle through Lynx.

Real-Time Rail

Payments Canada + Interac

ISO 20022 instant-payment system intended to add 24/7 irrevocable retail payments and richer data.

Target launch Q4 2026; designed to settle in central-bank money.

Securities FMI | TMX owns both the depository and domestic CCP

CDS / CDSX

TMX Group

Canadian CSD, clearing and settlement infrastructure for equities, debt and money-market instruments.

Securities move in CDSX; payment obligations settle through designated banks and Bank of Canada arrangements.

CCMS

TMX + Clearstream

Collateral inventory, mobilisation and optimisation service linking Canadian market participants.

It changes collateral selection and movement; it is not a payment system or digital Canadian dollar.

CDCC

TMX Group

CCP for listed derivatives, fixed-income transactions and selected equity products.

Margin, netting and default management are vertically integrated with the exchange group.

Public oversight

Bank of Canada + provincial regulators

The Bank designates and oversees systemically important systems; securities regulation remains provincial.

Ownership, operation, supervision and settlement asset are separate control layers.

Digital money | contingency research, then Agorá accession

CONTINGENCY RESEARCH

Retail CBDC

The Bank of Canada completed extensive design research and moved active development into a readiness programme; issuance would depend on future need and public authority.

Purpose: preserve the capacity to respond if cash use collapses or private digital money threatens monetary sovereignty.

JOINED MAY 2026

Project Agorá

Canada joined after the seven-currency prototype report, bringing the Bank of Canada and a new Canadian private-sector cohort into later exploration.

Purpose: test the two-tier cross-border model against Canada's Lynx-centred monetary architecture.

WHOLESALE PRECURSOR

Project Jasper

Earlier Bank of Canada experiments tested DLT wholesale payments, securities DvP and cross-border settlement.

Purpose: understand whether distributed systems improve existing institutional markets.

COLLATERAL INFRASTRUCTURE

CCMS

The live TMX/Clearstream service mobilises and optimises securities collateral; the Bank of Canada plans to use it for market operations.

Purpose: coordinate production collateral inventory and movement across participating institutions.

Eight stacks, one comparison

Area	Public cash anchor	Core securities FMI	Digital-money path	What is actually being settled
Euro	T2 / TIPS	T2S; Euroclear; Clearstream; CCPs	digital euro; Pontes; Appia	retail public money vs token-security cash leg
Japan	BOJ-NET	BOJ JGB; JASDEC; JSCC	BOJ CBDC pilot; DCJPY	retail CBDC research vs deposit-token DvP
Korea	BOK-Wire+	KSD; KRX CCP	Hangang; Agorá	deposit tokens over wholesale CBDC
Mexico	SPEI / SIAC	Indeval DALI; CCV; Asigna	CoDi; Agorá	account money vs cross-border token claims
Switzerland	SIC	SIX SIS; x-clear; SDX	Helvetia; RTGS link	tokenised bonds against wCBDC or SIC cash
UK	RT2	CREST; LCH; ICE; LME	digital pound; Fnl-ity; GBTD	retail CBDC research, system money, deposits
US	Fedwire / FedNow	DTC; NSCC/FICC; CME/OCC/ICE	TCH; Kinexys; stablecoins	private token dollars, no Fed CBDC token
Canada	Lynx	CDS; CDCC; CCMS	CBDC contingency; Agorá	wholesale claims; collateral optimisation

Sources: Central-bank and FMI sources listed on the preceding 24 jurisdiction slides.

PART II

Project Agora Principles

Tokenised deposits, central-bank money and the programmable transition

Seven questions organise the findings

REPORT REPORT STRUCTURE

Question answered	What the report contributes
Why Agorá?	vision, participants, objectives and explicit assumptions
What must improve?	pain points, 15 business requirements and design choices
What was built?	prototype architecture and end-to-end / PvP workflows
How did it work?	ledger, contracts, participant suite, privacy and testing
Could it be legal?	asset character, finality, compliance, data and rulebooks
Did it help?	three scenarios, qualitative benefits and limitations
What comes next?	outcomes and technical, functional and legal work

The report establishes prototype feasibility; production still requires an operator, rulebook, resilience regime and launch decision.

Sources: BIS, Project Agorá final report, 27 May 2026, contents and Executive Summary.

42 private participants: global and European banks

- BNP Paribas
- BNY
- Citi
- Crédit Agricole CIB
- Deutsche Bank
- HSBC
- JPMorgan Chase
- Standard Chartered
- UBS
- Santander
- BBVA
- CaixaBank
- Commerzbank
- Groupe BPCE
- Lloyds Banking Group
- NatWest Group
- FNBO
- TD Bank N.A.

Eighteen of the 42 private participants are global, European or US banks; the cohort spans issuers, correspondents and potential liquidity providers.

Sources: BIS, Project Agorá report, participant roster, May 2026.

42 private participants: regional banks

Japan

- Mizuho Bank
- MUFG Bank
- SBI Shinsei Bank
- SMBC

South Korea

- Hana Bank
- IBK
- KB Kookmin Bank
- NongHyup Bank
- Shinhan Bank
- Woori Bank

Switzerland

- AMINA Bank
- Banque Cantonale Vaudoise
- Basler Kantonalbank
- PostFinance
- Sygnum Bank

Latin America

- Banco BV
- Banorte
- Monex

Eighteen regional banks supply domestic account, compliance and currency expertise.

Sources: BIS, Project Agorá report, participant roster, May 2026.

42 private participants: FMIs and networks

Participant	Capability represented in the cohort
Eurex Clearing	CCP risk management, margin, default management and settlement dependencies
Euroclear	CSD/ICSD settlement, custody, collateral and cross-border securities links
SIX Digital Exchange	regulated DLT exchange and CSD operations
Mastercard	network rules, identity, risk controls and bank distribution
Visa	network rules, risk controls and tokenised-bank-liability initiatives
Swift	ISO 20022 connectivity, identity and a competing shared-ledger programme

Six infrastructure participants contribute clearing, custody, securities settlement, network and messaging capabilities.

Sources: BIS, Project Agorá report, participant roster, May 2026; participant disclosures.

The correspondent workflow

REPORT WHY CHANGE THE WORKFLOW

Why the current chain is costly

- Multiple ledgers update sequentially.
- Compliance and data checks repeat at each hop.
- Operating hours and cut-offs do not align.
- Liquidity is prefunded across correspondent accounts.
- Exceptions trigger bilateral investigation and reconciliation.

Sources: BIS Agora final report, Executive Summary and Chapter 1.

The unified-ledger proposition

- Put tokenised commercial-bank deposits and central-bank reserves into one coordinated workflow.
- Preserve two-tier banking and the singleness of money.
- Prepare payment legs and compliance in parallel.
- Commit the complete transaction, or cancel it.

Four objectives and hard boundaries

REPORT OBJECTIVES + BOUNDARIES

Four objectives

1. **Interoperability** across money and currency ledgers.
2. **Programmability and composability** of the workflow.
3. **Atomic settlement** across every required payment leg.
4. **Enhanced compliance** using richer, reusable outcomes.

Agorá established prototype feasibility. Systemic operation still requires live integration, resilience and market-scale evidence.

Sources: BIS Agorá final report, Chapter 1.

What the prototype assumed

- Each central bank retains account and access policy.
- Banks can fund tokens from existing systems.
- The platform is the golden source for token balances.
- Production scalability, cyber resilience, liquidity-saving, live FX and tokenised securities are out of scope.

Fifteen business requirements

REPORT BUSINESS REQUIREMENTS

Authority and money

R1–R3

Preserve issuer autonomy; support issuance, redemption and transfer; manage settlement risk atomically.

A shared workflow must not erase central-bank or commercial-bank control of its own liability.

Visibility and privacy

R10–R11

Give participants useful status and reporting while selectively sharing transaction data.

Common state need not mean universal data access.

Route and integrity

R4–R5

Find viable correspondent paths and preserve cross-currency integrity across all legs.

No partial value transfer when one currency leg fails.

Liquidity and governance

R12–R13

Monitor liquidity and define operational, issuer and participant roles.

The prototype tests balances; production needs institutions.

Access and compliance

R6–R7

Authorise only permitted actors; prevalidate sanctions, AML/KYC, fraud and payment data.

Checks remain institution-specific even when outcomes are shared.

Non-functional controls

R14

Reliability, security, scalability, performance and recoverability.

Required by the business specification but not performance-tested.

Standards and exceptions

R8–R9

Use ISO 20022 / CBPR+ / HVPS+; support investigation, repair, return and cancellation.

The happy path is not enough for production payments.

Finality

R15

Make completion legally and operationally irreversible in every affected jurisdiction.

Technical commit and legal finality must be aligned, not assumed.

Sources: BIS Agorá final report, Chapter 2, requirements 1–15.

Five design choices

REPORT DESIGN PRINCIPLES

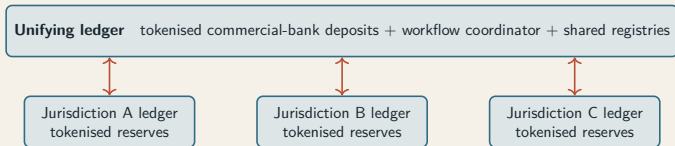
1. **Coordinate balance updates** across central-bank and commercial-bank ledgers instead of moving every claim onto one ledger.
2. **Confirm the payee** before value moves, reducing misdirection and unusable beneficiary data.
3. **Discover a path** through correspondents before instructing the full chain.
4. **Share compliance outcomes, not complete case files**; each institution keeps its own controls and evidential record.
5. **Lock first, then commit atomically**: each money issuer reserves its leg and delegates only the limited authority needed for settlement.

The innovation is coordination: information and liquidity are prepared in parallel, then a single workflow decides commit or cancel.

Sources: BIS Agorá final report, Chapter 2.

Two ledger layers preserve two-tier money

REPORT TWO-LAYER ARCHITECTURE



Participant nodes run ledger clients; asset, workflow and reference-data contracts govern permitted state changes.
Sensitive compliance and bank processing remain off-ledger.

The unified workflow preserves separate balance sheets: deposits remain claims on named banks and reserve tokens remain claims on named central banks.

Sources: BIS Agorá final report, Chapter 3.

The five-stage payment workflow

REPORT PAYMENT WORKFLOW

1. **Confirmation of payee:** payer and payee banks compare names and account data inside a temporary privacy group; the workflow retains the match outcome.
2. **Payment discovery and matching:** banks exchange bilateral capacity and conditions hop by hop; the search can backtrack until it finds an executable correspondent path.
3. **Validation:** every bank performs sanctions, AML/KYC, fraud, amount and operational-readiness checks, reporting only prescribed outcomes.
4. **Lock and delegate:** each issuer locks the required deposit or reserve balance and gives a smart contract narrowly scoped, temporary spending authority.
5. **Settle or cancel:** the coordinator verifies all locks, spends every leg, and completes the workflow; if prerequisites fail, locks are cancelled.

Agorá's atomicity is a workflow guarantee across ledgers. Legal finality requires a separate rulebook and statutory analysis.

Sources: BIS Agorá final report, Chapter 3.

The technical stack

REPORT TECHNICAL STACK

Ledger and consensus

- Hyperledger Besu, Ethereum Virtual Machine compatible.
- Permissioned validators using QBFT proof of authority.
- Deterministic contract execution and ledger ordering.

Asset functions

- Mint, burn and transfer.
- Create, delegate, spend and cancel a lock.

Sources: BIS Agorá final report, Chapter 4.

Workflow and reference contracts

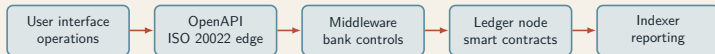
- Coordinator on the unifying ledger.
- Payment-leg contracts on each relevant ledger.
- Network, token, corridor and reserve-account registries.

Important qualification

- QBFT gives deterministic technical state.
- It does not by itself create legal settlement finality.

The participant suite

REPORT PARTICIPANT SUITE



pacs.008 and pacs.009 messages enter at the edge; middleware performs bank-specific checks and translates them into permitted ledger actions.

- Institutions can deploy components across cloud, multiple clouds or on-premises environments.
- Sensitive screening logic and case evidence stay within the bank's control boundary.
- The report leaves the production operator and hosting/governance model undecided.

Sources: BIS Agorá final report, Chapter 4.

Privacy, PvP and assurance

REPORT PRIVACY + PVP + ASSURANCE

Selective visibility

- Privacy groups isolate bilateral or limited-party processing.
- Confirmation-of-payee data can be processed ephemerally.
- The shared ledger stores only the state and outcomes required to advance the workflow.

PvP special case

- No payee confirmation, path search or customer amount is needed.
- Both currency legs validate and lock; the coordinator commits or cancels both.

Sources: BIS Agorá final report, Chapter 4.

What testing established

- Four milestone cycles and participant user-acceptance testing.
- Functional execution of selected scenarios.
- Qualitative assessment of the business requirements.

What it did not establish

- No stress, throughput or latency benchmark.
- No production cyber, failover or resilience test.
- No live RTGS or core-banking integration.

What the tokens are in law

REPORT LEGAL CHARACTER OF MONEY

- A reserve token is the definitive platform record of an underlying central-bank reserve-account balance; a deposit token records a named bank's deposit liability.
- The legal debtor-creditor relationship remains with the issuer. The token records that existing claim against a named bank or central bank.
- Minting and burning correspond to credits and debits of the relevant account; transfer changes the recorded account balance.
- Smart-contract code executes rights and duties established by the governing law, rulebook and participant agreements.
- Eligible direct actors are central banks and commercial banks; an FMI or CCP may initiate a PvP workflow. NBFIs and corporates participate indirectly through banks; natural persons are outside scope.

The prototype preserves familiar money law, but only if contracts and public law make the ledger entry authoritative.

Sources: BIS Agorá final report, Chapter 5.

The legal finality trigger

REPORT FINALITY

1. Each issuer first locks its own balance and delegates only the authority needed for the transaction.
2. Because the ledgers update asynchronously, the report identifies an **atomic settlement trigger** on the unifying ledger.
3. The production rulebook would need that trigger to mark irrevocability, finality and discharge in every participating jurisdiction.
4. Insolvency law and any payment-system designation must recognise that result; otherwise a technically complete transfer may remain legally contestable.
5. Choice of law, mandatory local law and public policy may still produce jurisdiction-specific differences.

Atomic execution is a software fact. Finality is a legal allocation of when nobody may unwind the payment.

Sources: BIS Agorá final report, Chapter 5, finality and choice-of-law analysis.

Compliance and data

REPORT COMPLIANCE + DATA

Financial-crime controls

- Every bank performs its own sanctions, AML/KYC and fraud checks.
- Three prescribed pass/fail outcomes govern whether the workflow advances.
- Failure flags can create confidentiality or tipping-off risks.
- Regulators may need evidence beyond a binary outcome.

Privacy and localisation

- Privacy groups reduce disclosure but do not eliminate personal data.
- Customer consent may be needed for payee checks and outcome sharing.
- Korea imposes strict localisation constraints; Mexico requires some domestic logs and accounting records.
- Supervisory access and cross-border transfer rules differ by jurisdiction.

Shared outcomes reduce repetition only if secrecy, evidential and localisation rules let institutions rely on them.

Sources: BIS Agorá final report, Chapter 5.

Three benefit scenarios

REPORT BENEFITS

Scenario	Problem	Agorá mechanism	Claimed benefit
Mexico to US urgent invoice	known correspondent path; time-sensitive US\$200k payment	parallel validation and liquidity locks across deposit and reserve tokens	shorter workflow, less reconciliation and lower settlement exposure
Switzerland to Korea corporate payment	route initially unknown; EUR used as vehicle currency	confirmation of payee, path discovery, validation and coordinated settlement	better route visibility and reduced manual outreach
UK to Japan liquidity rebalance	two correspondents exchange GBP and JPY	validate, lock both currency legs and execute PvP	removal of principal risk from asynchronous FX settlement

The chapter identifies plausible sources of value; production cost, realised savings and market-wide liquidity effects remain unquantified.

Sources: BIS Agorá final report, Chapter 6.

The prototype's limits

REPORT PROTOTYPE LIMITATIONS

Not production-tested

- Cybersecurity and adversarial resilience
- Throughput, latency and peak-volume stress
- Node, network and regional failover
- Operational monitoring and incident response
- RTGS, core-banking and live compliance integration

Not fully solved

- Liquidity-saving and queue management
- Fee transparency to end customers
- Exception handling and customer outreach
- Production governance and operator model
- Cross-jurisdiction legal finality and insolvency

A successful prototype can still leave the expensive parts of a systemic utility almost entirely ahead.

Sources: BIS Agorá final report, Chapter 6.

Six outcomes

REPORT OUTCOMES

1. Tokenisation can improve the **workflow around money**, not only represent balances differently.
2. Public-private collaboration is necessary because central banks issue the anchor while commercial banks own customer and compliance processes.
3. A shared ledger can preserve issuer autonomy through separate assets, permissions and jurisdictional domains.
4. Tokenisation does not inherently change the legal nature of reserves or deposits.
5. Shared infrastructure does not require every participant to see every data item.
6. Interoperability among bank deposit tokens is central to preserving singleness in a multi-bank system.

The result is a defensible two-tier architecture. Selection of a market utility requires a separate institutional decision.

Sources: BIS Agorá final report, Chapter 7.

Three future work programmes

REPORT FUTURE WORK

Non-functional

- cyber and key compromise
- resilience and failover
- performance and stress
- external-system integration
- zero-knowledge proofs and MPC

Functional

- coordinated compliance
- liquidity-saving mechanisms
- standards and interoperability
- multi-ledger deposit tokens
- netting, batching, PvP and DvP

Legal + institutional

- participation and permitted assets
- finality and insolvency
- accounting and prudential treatment
- exceptions, liability and loss
- operator, governance and disputes

Production requires simultaneous engineering of infrastructure, market rules and public authority.

Sources: BIS Agorá final report, Chapter 7.

Real-value testing

REPORT JULY 2026 EXTENSION

- Seventeen transaction scenarios and about 30 transactions were executed with real value in six currencies.
- Aggregate value was approximately CHF 800,000; individual transactions ranged from CHF 9,000 to CHF 125,000.
- Average initiation-to-settlement time was about 80 seconds, without live RTGS or core-banking integration.
- Twenty-eight private and public institutions participated; five central banks joined the tests. The New York Fed, Banxico and Bank of Canada did not.
- Further higher-volume testing was planned, with no commitment to a production platform.

Real-value transfers establish transactional capability; ownership, operation and credible exit remain institutional decisions.

Sources: BIS, Project Agorá real-value testing, 30 July 2026.

Production checklist

REPORT PRODUCTION DECISIONS

Question	Prototype answer	Production decision still required
Who issues money?	each central or commercial bank	unchanged, but admission and redemption rules need authority
Who operates ledgers?	project deployment	legal entity, hosting, validators, upgrades and recovery
Who admits participants?	project cohort	objective criteria, due process, suspension and appeal
When is payment final?	deterministic technical commit	rulebook trigger, designation and insolvency recognition
Who sees data?	privacy groups and limited outcomes	lawful basis, localisation, regulator access and retention
Who bears loss?	not tested	cyber loss, erroneous code, default, outage and liquidity allocation
How can a currency exit?	not tested	portability, fallback settlement and continuity without the shared layer

The findings make operator design, governance and exit arrangements urgent production decisions.

Sources: Synthesis from BIS Agorá final report, Chapters 1–7.

PART III

Agorá, Jura, Dunbar and mBridge

Tokenised deposits, central-bank money and the programmable transition

Four projects settle different objects

Project	Money represented	Transaction tested	What the evidence establishes
Agorá	tokenised commercial-bank deposits and central-bank reserves	cross-border correspondent payments, including FX legs	a two-tier workflow can discover a route, complete compliance checks, lock issuer liabilities and commit atomically
Jura	real-value EUR and CHF wholesale CBDC plus tokenised commercial paper	FX payment-versus-payment and securities delivery-versus-payment	central-bank money and securities can settle atomically across permissioned Corda subnetworks
Dunbar	hypothetical AUD, MYR, SGD and ZAR wholesale CBDCs	direct cross-border transfers on two prototype stacks	multi-central-bank access, governance and issuer autonomy can be expressed on a common platform
mBridge	direct central-bank-issued digital money in participating currencies	real-value cross-border payments and FX	central banks can validate a shared purpose-built ledger and operate an MVP with commercial-bank users

The projects differ first in the liability transferred: deposits plus reserves, wholesale CBDC plus securities, simulated wholesale CBDCs, or direct multi-CBDC money.

Sources: BIS Innovation Hub reports for Projects Agorá, Jura, Dunbar and mBridge.

Four projects allocate control differently

Agorá

TWO LEDGER LAYERS

Unifying deposit ledger plus currency-specific reserve ledgers; every issuer controls its own liability.

Production decision: operator, validators, admission, upgrades and credible exit.

Jura

CORDA SUBNETWORKS

Separate currency subnetworks and notaries; Banque de France and SNB signatures govern EUR and CHF.

Production decision: availability, recovery and control of software change.

Dunbar

COMMON MULTI-CBDC

One common platform implemented in both Corda and Quorum; central-bank nodes issue currencies and apply access policy.

Production decision: shared governance, non-resident access and dispute authority.

mBridge

CENTRAL-BANK VALIDATORS

A custom ledger operated through participating authorities' validating nodes and currency administration.

Production decision: legal framework, access, interoperability and durable governance.

Sources: BIS Project Agorá report (2026); Jura (2021); Dunbar (2022); mBridge project materials and 2024 handover.

Agorá: deposits and reserves

- **Participants:** seven founding currency areas, Canada in later exploration, and 42 private institutions in the report roster.
- **Assets:** named-bank deposit tokens on a unifying ledger and named-central-bank reserve tokens on jurisdiction ledgers.
- **Workflow:** confirm payee, discover correspondent path, exchange compliance outcomes, lock every liability, then commit or cancel the chain.
- **Evidence:** simulated prototype followed by limited real-value transactions in July 2026; live RTGS, core-banking and systemic-scale testing remain future work.

Agorá moves routing, compliance evidence and settlement coordination into shared programmable infrastructure.

Sources: BIS, Project Agorá findings report (May 2026) and real-value testing summary (July 2026).

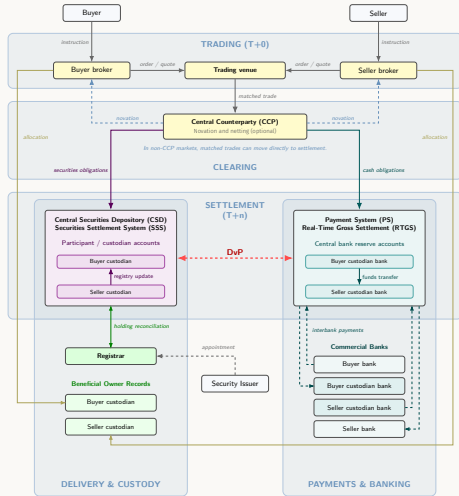
Jura: real money and tokenised securities

- **Participants:** Banque de France, SNB and BIS with SIX, UBS, Credit Suisse, Natixis, R3 and Accenture.
- **Assets:** real-value euro and Swiss-franc wholesale CBDC plus tokenised commercial paper.
- **Architecture:** separate Corda subnetworks; currency-specific notaries and dual signatures preserved each central bank's approval authority.
- **Evidence:** payment-versus-payment in EUR/CHF and delivery-versus-payment against securities, using intraday wholesale CBDC redeemed after the experiment.

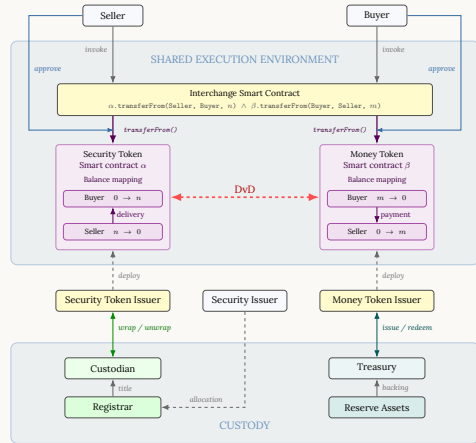
Jura demonstrated atomic FX and securities settlement; the platform operator controls availability, recovery and software evolution.

Sources: BIS, Project Jura report.

Delivery vs Payment



Delivery vs Delivery



Dunbar: common access to four wholesale CBDCs

- **Participants:** RBA, Bank Negara Malaysia, MAS and South African Reserve Bank, with banks and technology partners.
- **Assets:** hypothetical AUD, MYR, SGD and ZAR wholesale CBDCs; the experiment used simulated money.
- **Architecture:** two common-platform prototypes, one on Corda and one on Quorum, allowed direct transfers across participating banks.
- **Evidence:** technical flows were feasible; policy work concentrated on common rules, non-resident access, jurisdictional autonomy and governance.

Dunbar makes common access a constitutional choice: central banks must agree who may hold and transfer each currency.

Sources: BIS, Project Dunbar final report.

mBridge: central banks validate the shared ledger

- **Participants:** Hong Kong, Thailand, China and the UAE founded the project; Saudi Arabia joined in 2024; commercial banks executed real-value transactions.
- **Assets:** direct central-bank-issued digital money for domestic and cross-border use by eligible institutions.
- **Architecture:** a purpose-built mBridge Ledger with participating central banks operating validating nodes and administering their currencies.
- **Evidence:** real-value pilots, an MVP in 2024, and transfer of project stewardship from the BIS to participating authorities in October 2024.

mBridge assigns shared validation to monetary authorities; Agorá combines public reserve domains with a public–private deposit workflow.

Sources: BIS, Project mBridge.

Message-layer control is demonstrated power

- 2022: EU sanctions disconnected major Russian banks from Swift – the messaging layer used, openly, as a policy instrument.
- CIPS (2015) and the e-CNY are the most developed state-built responses; mBridge is the multi-central-bank variant.
- Every jurisdiction absorbed the same lesson: whoever operates the coordination layer can exclude.

The layers above final settlement carry decisive power. Their operator can shape admission, routing and continuity. documented

Sources: EU Council sanctions decisions (2022); PBoC CIPS materials; BIS mBridge documentation.

Where coordination authority sits

Model	Examples	Concentration danger
Common multi-issuer platform	Dunbar, mBridge	common governance and validator layer
Shared bank/public workflow	Agorá, Swift ledger	orchestration, identity and workflow-rule authority
Interlinked national ledgers	Cedar-Ubin+, Pontes, Meridian FX, Helvetia	bridge, oracle and synchroniser authority

Coordination authority can sit in one shared platform, a bank/public workflow layer, or the bridges between sovereign ledgers.

Sources: BIS Innovation Hub project reports; ECB Pontes; SNB Helvetia; New York Fed Cedar. Project-by-project control questions in the appendix.

PART IV

Swift and competing networks

Tokenised deposits, central-bank money and the programmable transition

Swift is moving from messages to commitments

- September 2025: shared-ledger plan designed with more than 30 institutions.
- March 2026: Besu-based MVP records and validates interbank payment commitments.
- Swift operates the ledger; banks retain keys, deposit assets, compliance and ultimate funding.
- July 2026: controlled initial-use phase announced with 17 banks.

Swift is positioning its ledger as the trusted workflow that records commitments while banks retain the assets, keys, compliance and ultimate funding.

Sources: Swift, shared-ledger announcements of September 2025, March 2026 and 9 July 2026.

The 17-bank cohort is globally distributed

Europe / Americas

- BNP Paribas
- BNY
- Citi
- HSBC
- Itaú
- Lloyds
- UBS
- Wells Fargo

Asia-Pacific

- ANZ
- DBS
- MUFG
- OCBC
- UOB

Middle East / Africa

- First Abu Dhabi Bank
- FirstRand
- Mashreq
- Standard Chartered

Sources: Swift, 9 July 2026 initial-use announcement.

JPMorgan is absent from 17, not from the strategy

- JPMorgan was in Swift's earlier design group, but not the July 2026 cohort of 17.
- It is an Agorá participant and owns **Kinexys** – now settling in eight currencies after a mid-2026 Asia-Pacific expansion (AUD, HKD, JPY, CNY, SGD added).
- **JPMD**, the renamed dollar deposit token, is live on Base; native Canton issuance was announced in January 2026.
- KB Kookmin – an Agorá and Han River bank – is adopting Kinexys for instant USD payments.

JPMorgan is hedging architectures, not choosing one.

Sources: Swift design and cohort releases; JPMorgan Kinexys milestones 2026; Digital Asset/Kinexys Canton announcement, 7 January 2026.

Four production comparators

CLS

FX PVP

Atomic FX settlement in 18 currencies; over US\$8tn daily; bank ownership, mature finality and cooperative central-bank oversight.

Agorá test: equivalent legal certainty, liquidity efficiency and operating resilience.

T2S

SECURITIES DVP

Cross-CSD delivery-versus-payment in central-bank money under Eurosystem operation and harmonised rules.

Agorá test: coordinate legal issuers while preserving authoritative records.

CHIPS

USD LIQUIDITY

High-value dollar clearing with continuous matching, netting, pre-funding, loss controls and Federal Reserve oversight.

Agorá test: queues, netting, credit controls and stress-tested liquidity.

Swift

GLOBAL WORKFLOW

Identity, secure messaging, ISO 20022, admission, PKI, service management and recovery at global scale.

Agorá test: production identity, bank/RTGS integration and exception handling.

Sources: CLSSettlement; ECB T2S; The Clearing House CHIPS; Swift service and governance materials.

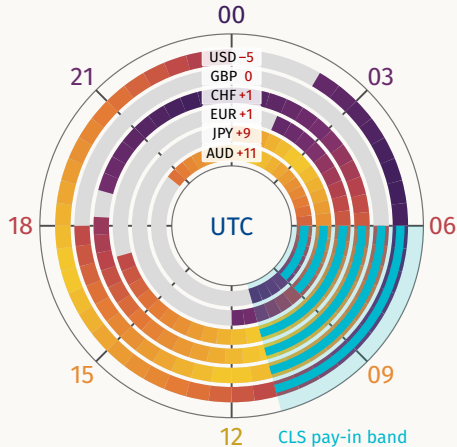
CLS sets the production benchmark for PvP

Dimension	CLSSettlement today	Agorá production test
Scope	final and irrevocable PvP settlement for eligible FX in 18 currencies	atomic multi-leg payments across deposit tokens and reserve domains
Liquidity	multilateral netting cuts funding requirements by over 96%; optional tools reduce them further	queueing, netting, batching, credit and stress liquidity remain future work
Access	75+ settlement members and 38,000+ indirect users through members	admission, suspension, indirect access and currency-specific eligibility need a rulebook
Governance	bank-owned utility, Federal Reserve supervision and cooperative oversight by eligible-currency central banks	legal operator, board, validator powers, change control and issuer vetoes need allocation
Operations	resilient production service linked to 18 RTGS systems, with defined funding and settlement cycles	live RTGS/core-banking integration, recovery and systemic-scale performance need testing

CLS benchmarks the PvP leg; Agorá additionally coordinates named-bank deposits, route discovery, compliance outcomes and currency-specific reserve domains.

Sources: CLSSettlement; BIS Project Agorá findings report.

Continuous Linked Settlement (PvP)

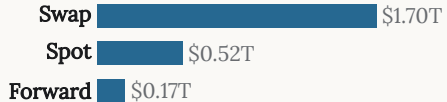


CLS dominates global PvP FX settlement in 18 currencies. Established 2002 to manage **Herstatt risk**.

~50% of global FX volume; netting efficiencies ~96%.

CLS average daily volume

Total: \$2.38T (Dec 2025)



TCH proposes bank-owned tokenised deposits

- 5 June 2026 proposal backed by 17 named institutions, from JPMorgan and Bank of America to Truist and Fifth Third.
- On-chain clearing and settlement of tokenised bank deposits, with a connectivity layer to CHIPS and RTP.
- TCH is owned by its large US member banks; technology provider still being selected.
- Press reporting points to a first-half 2027 launch for the network.

The bank branch of programmable dollars preserves deposit franchises and owner-bank governance.

Sources: The Clearing House, 5 June 2026 initiative.

Open USD proposes a distribution coalition

- Announced 30 June 2026 by the **Open Standard** coalition, led by Bridge founder Zach Abrams; more than 140 partners named, including Visa, Mastercard, Stripe, BlackRock and Coinbase.
- Not yet live – targeting a second-half 2026 launch; fee-free uncapped mint and redemption proposed; reserve earnings shared with participating distributors after a management fee.
- Context: the GENIUS Act (July 2025) bars paying yield to stablecoin holders – revenue-sharing with distributors is the compliant substitute.

Open USD competes by paying platforms for distribution; TCH competes by preserving bank deposits.

Sources: [Open Standard](#) announcement, 30 June 2026; CoinShares research note, July 2026; GENIUS Act (2025).

Open USD: disclosed architecture and open fields

- **Operator:** Open Standard describes an independent operating company governed by a board drawn from partners; press coverage describes the company as issuer. **documented**
- **Issuer and licence:** the incorporated issuer, charter and GENIUS-Act application remain undisclosed ahead of the proposed 2026 launch. **documented**
- **Mint/redeem:** fee-free and uncapped minting and redemption are proposed; direct eligibility, timing and the management fee remain open.
- **Chains:** reporting identifies Tempo and Solana for native issuance; an official chain list remains pending. **inference**
- **Reserves and partners:** reserves are described only as maintained at major financial institutions. Coinbase reaffirmed Circle; Visa and Mastercard described broad support for multiple coins. **documented**

Distribution is extensively described; issuer identity, reserves and governance remain the decisive open fields.

Sources: Open Standard announcement (30 June 2026); American Banker; Fortune; The Block; Forbes (all 30 June–2 July 2026).

Tokenised MMFs become payment-adjacent collateral

- BUIDL, BENJI/FOBXX and USYC combine fund claims with transferable tokens.
- They can accelerate subscription, redemption and collateral mobility.
- They are securities/fund interests—not bank deposits or central-bank money.
- NAV, gates, official ownership records and eligible-investor restrictions remain decisive.

A token can trade like cash operationally while retaining investment and insolvency risk legally.

Sources: Franklin Templeton BENJI; Circle USYC; Securitize/BlackRock BUIDL materials.

Identity is the quiet gatekeeper

Architectures

- Central registry: one provider owns eligibility data
- Reusable issuer-signed credentials
- Federated multi-authority verification
- Zero-knowledge proofs of eligibility

Securitize, the live instantiation

- Transfer agent, broker-dealer/ATS, fund admin
- Allowlists encode jurisdiction and eligibility
- BlackRock: 2024 round and a board seat
- Listed as SECZ, July 2026 (Cantor-sponsored)

Whoever controls the identity-to-wallet mapping can become the private gatekeeper to tokenised markets.

Sources: W3C Verifiable Credentials 2.0; Securitize corporate and investor releases; SEC filings.

PART V

Technology and company lineage

Tokenised deposits, central-bank money and the programmable transition

Technology choice is constitutional choice

	Data model	Strongest feature	Capture point
Besu Canton	shared EVM state party-specific contracts	standards and labour pool privacy and cross-app atom- icity	validators, privacy, upgrades synchroniser, validators, Daml
Corda Fabric	shared bilateral states channels / private data	regulated workflows modular enterprise control	notary and network manager ordering and membership service
Public chain	replicated state	reach and composability	validators, bridges, RPC, custody

Sources: LF Decentralized Trust; Digital Asset; R3; Ethereum documentation.

Besu is the institutional common denominator

- Apache-2.0 Ethereum client under LF Decentralized Trust.
- Public Ethereum or permissioned QBFT/IBFT networks.
- EVM compatibility exposes mature tools and token standards.
- Used differently by Agorá, Swift, Fnality, DTCC and JSCC.

Open code reduces code lock-in; it does not eliminate control over validators, hosting, privacy services, keys or releases.

Sources: LF Decentralized Trust, [Hyperledger Besu](#); operator disclosures.

Canton coordinates private applications

- Participants hold party-specific contract views rather than one public payload ledger.
- Synchronisers sequence encrypted messages; mediators coordinate atomic commit.
- Global Synchronizer governance uses Foundation and super-validator voting.
- No major new FMI has chosen Corda since about 2022; Canton is where the regulated-DLT market consolidated.

Canton distributes data but can still concentrate operational power in synchronisation, validation, hosting and Daml expertise.

Sources: Digital Asset Canton architecture; Canton Foundation governance; companion taxonomy.

Institutions hedge every architecture

- JPMorgan: Agorá, Swift design, TCH, Kinexys, Partior and Canton.
- DTCC: Digital Asset investor; Canton and Besu deployments.
- Euroclear: Agorá, Fnality and Canton collateral network.
- Visa and Mastercard: Agorá, Swift/Besu groups and Open USD.
- Tradeweb: Digital Asset and Securitize; majority owned by LSEG.

Strategic optionality lets incumbent institutions shape the winner whichever stack prevails.

Sources: Corporate investment, participant and partnership disclosures mapped in the companion report.

PART VI

Australia, briefly

Tokenised deposits, central-bank money and the programmable transition

Australia: the core is owned, the layers above are not

- RBA owns and operates **RITS**; FSS supplies 24/7 settlement for NPP; CHES remains the legal-title spine.
- Above the final ledger: Swift messaging **L6**, TNS network operations **L6**, TCS BaNCS for the CHES replacement **L5**, undecided future orchestration **L2 L7**.
- The largest concession already happened: the large majority of AUD OTC interest-rate derivatives clears at **LCH SwapClear** in London, under CFR safeguards. **documented**

Australia issues, records and finally settles its own currency. It does not control the path end to end – and has not for a decade.

Sources: RBA RITS/FSS; AusPayNet; ASX; TCS; CFR OTC derivatives reports; LSEG SwapClear.

ASX: failure, conflict and consequence

- 2016: Digital Asset stake and board seat; 2017: DLT commitment; November 2022: pause and A\$245–255m write-off; 2023: TCS BaNCS selected.
- 20 April 2026: Release 1 (clearing) live on TCS BaNCS, ASX's first production workload on public cloud; Release 2 (settlement and subregister) targeted for 2029.
- ASIC inquiry (April 2026): resilience compromised in pursuit of high shareholder returns; A\$150m capital charge; independent clearing and settlement boards; revised transformation plan agreed with ASIC and the RBA on 1 July 2026. **documented**
- July 2026: A\$20.5 million Federal Court penalty for the misleading “progressing well” statement of February 2022, plus A\$3 million costs. **documented**

Vendor selection, venture investment, product development and critical-system replacement became inseparable. The claim that Clearstream “took CHESS” remains unproven.

Sources: ASX announcements; ASIC inquiry findings and penalty.

Acacia has piloted the future stack

- **Project Acacia** (RBA/DFCRC): 24 use cases selected, **20 completed** August 2025–February 2026 – 12 real-money pilots and 8 proofs of concept; about A\$4.4m of pilot wholesale CBDC issued across three platforms.
- Final report published **May 2026**: estimated economic gains of roughly A\$24 billion a year; next steps include a digital-FMI regulatory sandbox, RITS upgrades and an expanded Deposit Token Working Group.
- Flagship repo use case (18 May 2026): CBA–J.P. Morgan (Kinexys)–ASX–HQLAx, with Deutsche Börse as HQLAx's trusted third party. **Repo settled using a CBA deposit token** alongside pilot wCBDC; tokenised assets held at ASX.

The domestic pilot already contains every element of this briefing: commercial-bank settlement money, a US bank's platform and a Deutsche Börse-linked collateral layer above local assets. **documented**

Sources: RBA, Acacia final report release and report PDF (May 2026); CBA, 18 May 2026; J.P. Morgan, Kinexys trial.

What any jurisdiction should demand

1. The central bank can inspect and reproduce every material decision.
2. Liquidity locks are participant-authorised, capped and revocable in emergency.
3. A direct route exists that bypasses the optimiser.
4. Another operator can substitute without data loss.
5. The jurisdiction retains keys, telemetry and recovery authority.
6. The currency domain can isolate and continue domestically.
7. Net obligations in the national currency are finally discharged only in central-bank money, unless expressly authorised otherwise.

The machinery exists in current law. Exit that has never been rehearsed is not an exit plan.

The conclusion is adversarial, not conspiratorial

Retaining the last ledger entry is insufficient if private institutions control every condition that precedes it.

- Assume every actor will exploit lawful control and switching costs.
- Demand transparent optimisation and public emergency authority.
- Preserve multiple routes, domestic competence and credible substitution.
- Distinguish documented conflict from unproved coordination.

Sources: Project Agorá integrated briefing, 5 August 2026.

APPENDIX

Participants, lineage, ownership and sources



Reference material

Digital Asset: lineage in brief

- Founded 2014–15 out of the DRW orbit (Wilson/Hirani financial founders; Rooz, Saraniecki and Kfir the operating-technical group; Blythe Masters CEO from launch). **inference**
- Technology assembled by acquisition: Hyperledger Inc., Bits of Proof, Blockstack.io, Elevance (whose contract language became Daml); the Hyperledger mark was donated to the Linux Foundation.
- 2016: US\$50m+ from banks, exchanges and CSDs – ASX among them, with a board seat; 2025: US\$135m (Tradeweb/DRW); 11 June 2026: US\$355m led by a16z with ADIA, at a valuation above US\$2 billion; 700+ Canton ecosystem participants.
- Canton should not be projected backwards into the platform ASX selected in 2016.

Sources: Digital Asset launch, financing and corporate-history releases; PR Newswire, 11 June 2026.

Other projects test different control points

Project	Mechanism	Control question
Mariana Cedar–Ubin+	AMM for wCBDC FX atomic links across separate ledgers	who sets curves and supplies liquidity? can interoperability avoid a central platform?
Helvetia	live wCBDC versus RTGS link	integrated token or synchronised conventional money?
Pontes	DLT-to-TARGET bridge	how does market DLT reach public money?
Meridian FX	external queue and liquidity saving	can an optimiser shape settlement before RTGS?

Sources: BIS Innovation Hub project reports; ECB Pontes; SNB Helvetia; New York Fed Cedar.

Primary sources

- BIS – Project Agorá findings report (27 May 2026)
- BIS – Agorá real-value testing (30 July 2026)
- Swift – 17-bank ledger cohort (9 July 2026)
- TCH – bank-led on-chain money (5 June 2026)
- Open Standard – Introducing Open USD (30 June 2026)
- DCP Co. – DCJPY
- RBA/DFCRC – Project Acacia final report (May 2026)
- ASIC – CHESS penalty (July 2026)
- BIS – Project Dunbar · Jura · mBridge
- ECB – Pontes · SNB – Helvetia